Heart rate variability stabilization in athletes: towards more convenient data acquisition

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Summary

Resting heart rate variability (HRV) is a potentially useful marker to consider for monitoring training status in athletes. However, traditional HRV data collection methodology requires a 5-min recording period preceded by a 5-min stabilization period. This lengthy process may limit HRV monitoring in the field due to time constraints and high compliance demands of athletes. Investigation into more practical methodology for HRV data acquisitions is required. The aim of this study was to determine the time course for stabilization of ECG-derived lnRMSSD from traditional HRV recordings. Ten-minute supine ECG measures were obtained in ten male and ten female collegiate cross-country athletes. The first 5 min for each ECG was separately analysed in successive 1-min intervals as follows: minutes 0–1 ($lnRMSSD_{0-1}$), 1–2 ($lnRMSSD_{1-2}$), 2–3 ($lnRMSSD_{2-3}$), 3–4 ($lnRMSSD_{3-4}$) and 4–5 ($lnRMSSD_{4-5}$). Each 1-min lnRMSSD segment was then sequentially compared to lnRMSSD of the 5- to 10-min ECG segment, which was considered the criterion ($lnRMSSD_{Criterion}$). There were no significant differences between each 1-min lnRMSSD segment and $lnRMSSD_{Criterion}$, and the effect sizes were considered trivial (ES ranged from 0·07 to 0·12). In addition, the ICC for each 1-min segment compared to the criterion was near perfect (ICC values ranged from 0·92 to 0·97). The limits of agreement between the prerecording values and $lnRMSSD_{Criterion}$ ranged from ±0·28 to ±0·45 ms. These results lend support to shorter, more convenient ECG recording procedures for lnRMSSD assessment in athletes by reducing the prerecording stabilization period to 1 min.

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Introduction

The utilization of heart rate variability (HRV) has grown considerably in the field of sports science (Buchheit, 2014). This non-invasive, cardiac-autonomic marker has been shown to reflect training load (Iellamo *et al.*, 2004; Edmonds *et al.*, 2013; Sartor *et al.*, 2013) and overreaching (Tian *et al.*, 2012; Le Meur *et al.*, 2013; Plews *et al.*, 2013) in competitive athletes. Others have found HRV to be useful in modulating training prescription on a day to day basis (Kiviniemi *et al.*, 2007, 2010; Botek *et al.*, 2013b). Further, there is evidence to suggest that athletic performance may be related to HRV in endurance athletes (Manzi *et al.*, 2009; Chalencon *et al.*, 2012; Flatt & Esco, 2014). Sport scientists therefore utilize HRV as an objective physiological marker to assist in training management as a means to optimize adaptation and performance in athletes (Plews *et al.*, 2013; Stanley *et al.* 2013; Buchheit, 2014).

The preferred HRV parameter for athlete monitoring is the vagally mediated, log-transformed root mean square of successive R-R interval differences (lnRMSSD) (Plews *et al.*, 2013; Buchheit, 2014). lnRMSSD has been shown to be a suitable HRV index for field assessment in athletes as it is less influenced by breathing rate compared to frequency domain measures (Penttilä *et al.*, 2001; Saboul *et al.*, 2013) and can be accurately assessed in 60 s or less (Esco & Flatt, 2014). Additionally, lnRMSSD may provide better stability and repeatability compared to spectral HRV markers (McNames & Aboy, 2006; Young & Leicht, 2011).

Unfortunately, traditional HRV methodology (Task Force of the European Society of Cardiology & the North American Society of Pacing & Electrophysiology, 1996) requires lengthy recording time periods that may limit longitudinal monitoring due to the difficulty of maintaining compliance from athletes (Botek *et al.*, 2013a; Sartor *et al.*, 2013; Plews *et al.*, 2014). For example, Young & Leicht (2011) demonstrated that a 10-min

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ECG recording provides a sufficiently stationary signal and has been the recommended recording duration for athletes (Aubert *et al.*, 2003). Within the literature, the suggested 10-min ECG period is often divided into a 5-min stability period followed by a 5-min HRV recording period. However, some researchers have used <5 min (Kiviniemi *et al.*, 2007; Plews *et al.*, 2012, 2014; Le Meur *et al.*, 2013; Sartor *et al.*, 2013), while others have used >5 min (Leicht *et al.*, 2003; Bara-Filho *et al.*, 2013; Botek *et al.*, 2013b) for stabilization prior to HRV assessment. Therefore, there remains a wide discrepancy within sports science literature as it pertains to the allotted time frames provided for stabilization preceding the criterion segment. Determining the time course for lnRMSSD stabilization in athletes is of particular importance for sport physiologists who wish to include HRV in their monitoring protocols. As no specific guidelines exist, further research in this area is warranted (Botek *et al.*, 2013a; Young and Leicht, 2011; Esco & Flatt, 2014).

The aim of this study was to determine the time course for stabilization of lnRMSSD over the first 5 min of a 10-min supine ECG recording in collegiate endurance athletes. As aerobic training leads to the improvement in one's ability to reach a period of physiological stasis and previous demonstrations of high RMSSD stability across longer durations in non-athletes (Young & Leicht, 2011), it is reasonable to hypothesize that stabilization of lnRMSSD may be achieved in <5 min among the current sample of collegiate endurance athletes.

Methods

Study population

Ten male and ten female endurance athletes from the University's National Association for Inter-Collegiate Athletics Varsity Cross Country team volunteered for this study. Descriptive statistics are shown in Table 1. Each subject provided written informed consent as approved by the Institutional Review Board for Human Participants. In addition, health-history questionnaires were completed which indicated that each athlete was apparently healthy, free from cardiopulmonary, metabolic and orthopaedic disorders. Data collection for each subject occurred 1 week prior to the start of the 2013 cross-country competitive season during the morning closely after awakening between the hours of 7:00 and 9:00 a.m. Each subject was to report to the laboratory in a fasted state, which involved the abstinence of all food following dinner and to avoid fluid intake after waking. Subjects were asked to refrain from the consumption of stimulants (e.g. caffeine) or depressants (e.g. alcohol) and to avoid strenuous exercise for 24 h prior to data collection.

Table 1 Descriptive statistics (mean ± SD) of the studied sample.

	Men (n = 10)	Women (n = 10)	All (n = 20)
Age (year)	19·8 ± 1·6	20·9 ± 1·8	20·4 ± 1·7
Height (cm)	176·4 ± 6·3	165·1 ± 7·8*	170·1 ± 9
Weight (kg)	67·1 ± 6·4	58·4 ± 7·5*	62·3 ± 8·1
Body fat (%)	10·4 ± 2·8	25·1 ± 4·2*	18·5 ± 8·3
VO_{2Max} (ml kg^{-1} min^{-1})	58·8 ± 6·4	47·0 ± 1·9*	52·2 ± 7·5

Values marked with an asterisk indicate a significant difference from men ($P \leq 0{\cdot}05$).

Procedures

Upon arrival at the Human Performance Laboratory, height and weight were recorded for each subject. Body mass was weighed on a calibrated digital scale (BWB-800AS; Tanita Corporation, Arlington Heights, IL, USA), and height was measured via wall mounted stadiometer (SECA 220; Seca, Ltd, Hamburg, Germany). Body composition and maximal oxygen consumption (VO_{2max}) were assessed for descriptive purposes following the resting ECG measures to preserve the resting condition of the athletes. Body fat percentage was determined with a calibrated dual-energy X-ray absorptiometer (GE Lunar Prodigy, Software version 10.50.086; GE Lunar Corporation, Madison, WI, USA). VO_{2max} was determined from a maximal graded exercise test on a Trackmaster treadmill (Full Vision, Inc., Carrollton, TX, USA), while expired gas fractions were analysed from the mouth with a metabolic cart (ParvoMedics TrueOne® 2400 metabolic cart, Sandy, UT, USA). A Polar heart rate monitor (Oulo, Finland) was used to measure exercise heart rate. Criteria for VO_{2max} determination included any two of the following: a plateau in oxygen consumption (±2 ml kg^{-1} min^{-1}) despite an increased work rate, respiratory exchange ratio of >1·15, heart rate within ten beats of age-predicted maximum (220-age) and volitional fatigue. All descriptive data are presented in Table 1.

For ECG analysis, each subject was brought into a quiet, dimly lit and climate controlled (approximately 20·5°C with 59% humidity) examination room. Subjects were instructed to rest in a supine position, while remaining as motionless as possible on a comfortable examination bed for the duration of the ECG recording. Once the subject assumed the supine position on the examination bed, a researcher placed surface electrodes on the subject in a lead II configuration. This involved the placement of three surface electrodes on the subject and was completed within 1 min to limit the length of supine rest prior to ECG recording. The electrodes were connected to a Biopac MP100 data acquisition system (Goletta, CA, USA), which was synched to a personal computer. AcqKnowledge software (v 3.9; Biopac) was used for collection of real-time ECG. Upon completion of ECG preparation (i.e. 1 min) data recording commenced and the subject was left alone at rest in a supine position for at least 10 min.

Once data collection was completed, all ECG recordings were transferred to Nevrokard software (version 11.0.2; Izola, Slovenia) for HRV analysis. After manual inspection of each ECG recording, ectopic beats and artefacts were removed and replaced by the adjacent normal cycle. More than three ectopic

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beats found within an ECG segment was considered as criteria for exclusion ($n = 0$). With the Nevrokard software, ECG data were converted to a tachogram for each subject where successive R-R intervals were displayed along the *y*-axis and, the number of beats within the ECG segment was displayed on the *x*-axis. From the tachogram, the root mean square of successive R-R interval differences was determined and transferred to a spreadsheet (SPSS/PASW software, version 18.0) where log transformations were applied.

In this study, the lnRMSSD of the 5- to 10-min segments were recorded and considered as the criterion reference ($lnRMSSD_{Criterion}$) for reasons described previously. Thereafter, lnRMSSD was analysed individually for minutes 0–1 ($lnRMSSD_{0–1}$), 1–2 ($lnRMSSD_{1–2}$), 2–3 ($lnRMSSD_{2–3}$), 3–4 ($lnRMSSD_{3–4}$) and 4–5 ($lnRMSSD_{4–5}$) of the prerecording 5-min ECG segment. The individual 1-min segments were then compared to the criterion in sequence to determine whether lnRMSSD achieved stability (i.e. showed no difference to the criterion) earlier than 5 min with the statistical procedures listed below. There was no significant difference ($P = 0{\cdot}68$) in $lnRMSSD_{Criterion}$ between the male ($lnRMSSD_{Criterion} = 4{\cdot}22 \pm 0{\cdot}62$ ms) and female ($lnRMSSD_{Criterion} = 4{\cdot}34 \pm 0{\cdot}56$ ms) athletes. Therefore, the lnRMSSD variables were compared across the entire sample as a whole ($n = 20$).

Statistical analyses

Statistical analyses were performed using SPSS/PASW (version 18.0; IBM Corporation, New York, NY, USA) and Microsoft Excel 2007 (Microsoft Windows Corporation, Redmond, WA, USA) software. Means and standard deviations (SD) were determined for all the studied variables. A Shipiro–Wilk test determined that the assumption of normality was met for all lnRMSSD values ($P>0{\cdot}05$). The difference between the lnRMSSD segments (i.e. $lnRMSSD_{0–1}$, $lnRMSSD_{1–2}$, $lnRMSSD_{2–3}$, $lnRMSSD_{3–4}$, $lnRMSSD_{4–5}$ and $lnRMSSD_{Criterion}$) was examined with a one-way repeated measures analysis of variance. Bonferroni follow-up tests were used to further explore the differences. Cohen's *d* was calculated to determine the effect size of the mean differences (Cohen, 1988) between the 1-min prerecorded values and $lnRMSSD_{Criterion}$ with thresholds considered as trivial (<0·2), small (0·2–0·6), moderate (0·6–1·2), large (1·2–2·0) or very large (>2·0) (Hopkins *et al.*, 2009). Intraclass correlations (ICC) for absolute agreement were used to evaluate the agreement between the lnRMSSD parameters. An ICC value between 0 and 0·30 was considered small; 0·31–0·49, moderate; 0·50–0·69, large; 0·70–0·89, very large; and 0·90–1·00, near perfect (Hopkins, 2009). The typical error was also determined for each ultra-short-term segment. In addition, Bland–Altman plots were formed to identify the upper and lower limits of agreement: that is, ±1·96 SD of the mean difference/bias between the criterion and ultra-short lnRMSSD values (Bland & Altman, 1986). *A priori* statistical significance was determined as $P<0{\cdot}05$.

Results

Table 2 provides the comparison statistics for the 1-min prerecording lnRMSSD segments to the $lnRMSSD_{Criterion}$. There were no significant differences between each 1-min lnRMSSD segment and $lnRMSSD_{Criterion}$, and the effect sizes were considered trivial (Cohen's *d* ranged from 0·07 to 0·12). In addition, the ICC for each 1-min segment compared to the criterion was considered near perfect (ICC values ranged from 0·92 to 0·97). Bland–Altman plots of the differences between the prerecording lnRMSSD values and criterion compared to the mean values are represented in Figs 1–5. The limits of agreement (±1·96 SD of the Bias) between the prerecording values and $lnRMSSD_{Criterion}$ ranged from ±0·28 to ±0·45 ms. A bar graph displaying means and standard deviations for each lnRMSSD segment in comparison to the criterion can be viewed in Fig. 6.

Discussion

To enhance the convenience of HRV monitoring for sport physiologists, shortened prerecording measurement procedures are desired but currently lack investigation. The aim of the current study was to determine the time course for stabilization of ECG-

Table 2 Comparison of the lnRMSSD (ms) values that were analysed individually for minutes 0–1 ($lnRMSSD_{0–1}$), 1–2 ($lnRMSSD_{1–2}$), 2–3 ($lnRMSSD_{2–3}$), 3–4 ($lnRMSSD_{3–4}$) and 4–5 ($lnRMSSD_{4–5}$) of the prerecording 5-min electrocardiograph segment to the criterion segment recorded between 5 and 10 min ($lnRMSSD_{Criterion}$).

						95% CI		Limits of agreement
	Mean ± SD	*P*	Cohen's *d*	Typical error	ICC	Lower	Upper	Bias ± 1·96 SD
$lnRMSSD_{Criterion}$	4·66 ± 0·39	–	–	–	–	–	–	–
$lnRMSSD_{0–1}$	4·63 ± 0·44	0·99	0·07	0·16	0·92	0·80	0·97	−0·03 ± 0·45
$lnRMSSD_{1–2}$	4·64 ± 0·45	0·99	0·05	0·12	0·96	0·89	0·98	−0·02 ± 0·34
$lnRMSSD_{2–3}$	4·62 ± 0·45	0·99	0·10	0·16	0·95	0·88	0·98	−0·04 ± 0·35
$lnRMSSD_{3–4}$	4·66 ± 0·47	0·99	0·00	0·14	0·95	0·87	0·98	−0·00 ± 0·39
$lnRMSSD_{4–5}$	4·61 ± 0·46	0·99	0·12	0·10	0·97	0·92	0·99	−0·05 ± 0·28

P value refers to the differences compared to the $lnRMSSD_{Criterion}$, ICC = intraclass correlation coefficient, CI = 95% confidence intervals of ICC, Bias = mean difference between the criterion (i.e. $lnRMSSD_{Criterion}$) and prerecording (i.e. $lnRMSSD_{0–1}$, $lnRMSSD_{1–2}$, $lnRMSSD_{2–3}$, $lnRMSSD_{3–4}$ and $lnRMSSD_{4–5}$) values.

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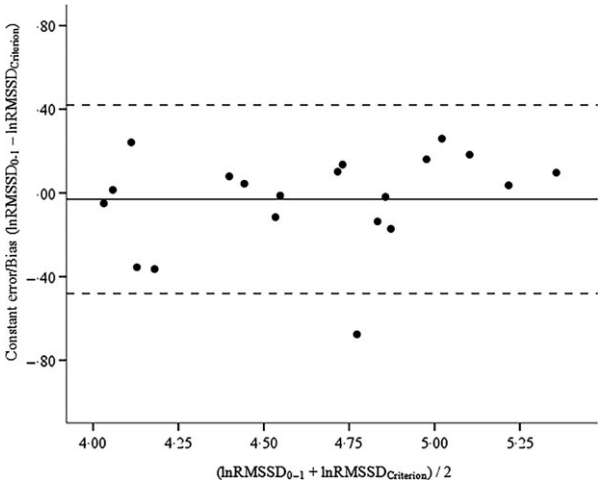


Figure 1 Bland–Altman Plot representing the limits of agreement between the criterion lnRMSSD of the 5- to 10-min segments ($lnRMSSD_{Criterion}$), and lnRMSSD that was analysed between minutes 0 and 1 ($lnRMSSD_{0–1}$) of the prerecording 5-min ECG segment. The solid middle line indicates the constant error, while the two dashed lines represent the upper and lower limits of agreement.

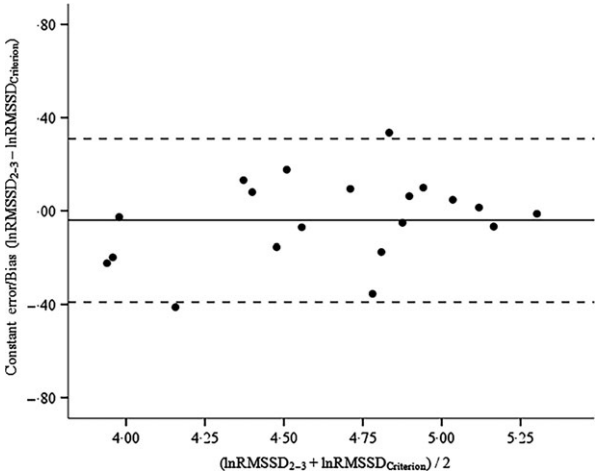


Figure 3 Bland–Altman Plot representing the limits of agreement between the criterion lnRMSSD of the 5- to 10-min segments ($lnRMSSD_{Criterion}$), and lnRMSSD that was analysed between minutes 2 and 3 ($lnRMSSD_{2–3}$) of the prerecording 5-min ECG segment. The solid middle line indicates the constant error, while the two dashed lines represent the upper and lower limits of agreement.

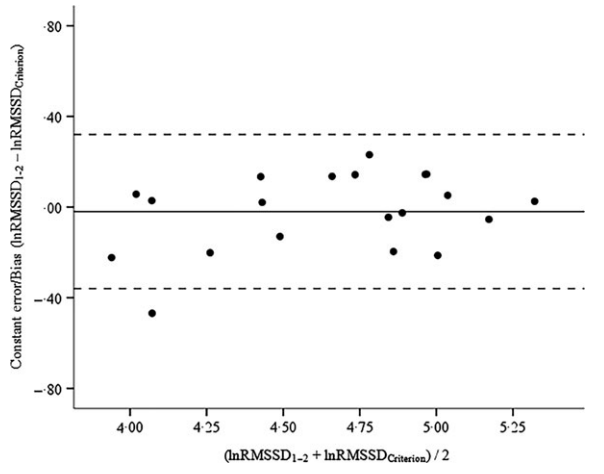


Figure 2 Bland–Altman Plot representing the limits of agreement between the criterion lnRMSSD of the 5- to 10-min segments ($lnRMSSD_{Criterion}$), and lnRMSSD that was analysed between minutes 1 and 2 ($lnRMSSD_{1–2}$) of the prerecording 5-min ECG segment. The solid middle line indicates the constant error, while the two dashed lines represent the upper and lower limits of agreement.

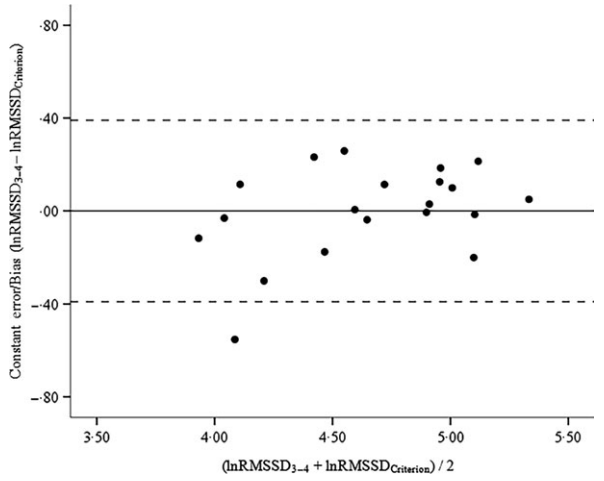


Figure 4 Bland–Altman Plot representing the limits of agreement between the criterion lnRMSSD of the 5- to 10-min segments ($lnRMSSD_{Criterion}$), and lnRMSSD that was analysed between minutes 3 and 4 ($lnRMSSD_{3–4}$) of the prerecording 5-min ECG segment. The solid middle line indicates the constant error while the two dashed lines represent the upper and lower limits of agreement.

derived lnRMSSD from within the traditional 5-min resting period that precedes standard 5-min recordings in collegiate endurance athletes. The results demonstrated no significant differences between each isolated 1-min segment of the prerecording stabilization period (i.e. $lnRMSSD_{0–1}$, $lnRMSSD_{1–2}$, $lnRMSSD_{2–3}$, $lnRMSSD_{3–4}$ and $lnRMSSD_{4–5}$) and $lnRMSSD_{Criterion}$. In addition, ICCs of the 1-min lnRMSSD segments when compared to the criterion were near perfect. Furthermore, Bland–Altman plots showed tight limits of agreement (±1·96 SD ranging from 0·28 to 0·45) for each of the 1-min prerecording lnRMSSD measures. Therefore, lnRMSSD measured via 1-min segments within the first 5 min of supine HRV recording appear to strongly agree with the $lnRMSSD_{Criterion}$. These results suggest that stabilization of lnRMSSD occurs within the first min of a 5-min prerecording period.

To our knowledge, no other study has specifically investigated the time course for stability of lnRMSSD among athletes, hence making comparisons difficult. However, investigations examining HRV stability before postural change and across longer durations have been previously performed in non-athletic subjects who may take longer to achieve a stable heart rate compared to aerobically trained individuals. For example, Botek *et al.* (2013a) found that orthoclinostatic (i.e. supine-standing-supine) HRV values (which did not include

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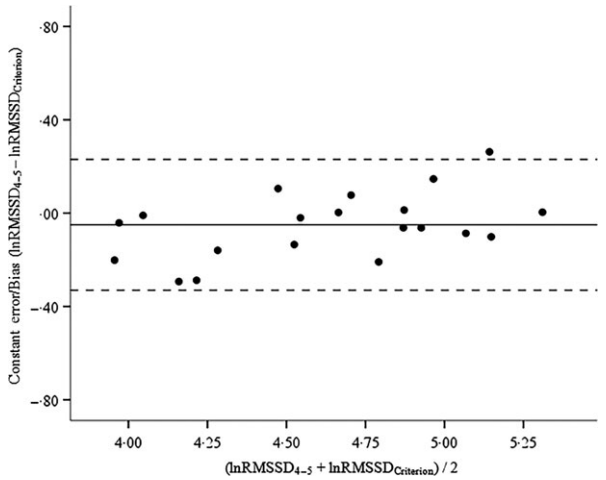


Figure 5 Bland–Altman Plot representing the limits of agreement between the criterion lnRMSSD of the 5- to 10-min segments ($lnRMSSD_{Criterion}$) and lnRMSSD that was analysed between minutes 4 and 5 ($lnRMSSD_{4–5}$) of the prerecording 5-min ECG segment. The solid middle line indicates the constant error, while the two dashed lines represent the upper and lower limits of agreement.

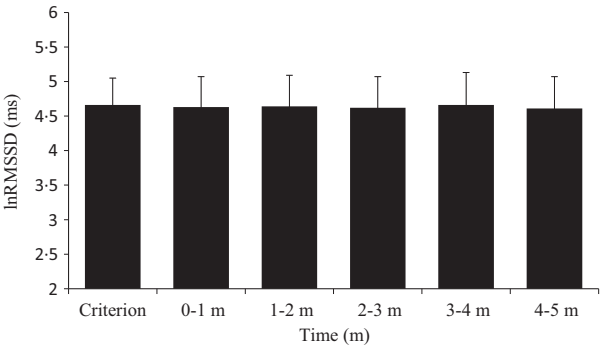


Figure 6 Bar graph representing the group mean and standard deviation for lnRMSSD at each successive 1-min interval in comparison to $lnRMSSD_{Criterion}$ (5–10 m).

lnRMSSD) were not significantly different when preceded by either a 1-min or traditional 5-min stabilization period. Young & Leicht (2011) determined that vagally mediated HRV parameters (including raw RMSSD) that were recorded during four consecutive 10-min ECG segments measured across 40 min were not significantly different in healthy non-athletic adults when analysed in three different positions: that is, supine, seated and standing positions. However, the orthostatic test performed by Botek *et al.* (2013a) is difficult to apply in sportive field settings and shortened time segments (i.e. 1-min intervals) were not evaluated in Young & Leicht (2011). Furthermore, neither of the two studies above included endurance athletes (Young & Leicht, 2011; Botek *et al.*, 2013a). It is well accepted that aerobically fit individuals possess greater resting vagal modulation and lower resting heart rates compared to sedentary controls (Dixon *et al.*, 1992), necessitating the current investigation within this particular population (Botek *et al.*, 2013a). The novel finding of the current study is that lnRMSSD assessed in consecutive 1-min segments over a traditional 5-min stabilization period showed no difference to $lnRMSSD_{Criterion}$ assessed under laboratory-controlled settings in collegiate endurance athletes. This data lend initial support to shorter, more convenient HRV data collection in athletes by reducing the stabilization period to 1 min.

Also relevant to the current study are recent investigations demonstrating that randomly selected 60-s lnRMSSD segments within a standard 5-min ECG recording period were no different from $lnRMSSD_{Criterion}$ among collegiate athletes (Esco & Flatt, 2014) and non-athletes (Nussinovitch *et al.*, 2011, 2012). However, these analyses were from ECG segments that were preceded by a 5-min stabilization period. Research from previous investigations (Nussinovitch *et al.*, 2011, 2012; Esco & Flatt, 2014) and the current investigation together suggest that lnRMSSD may be measured with a 1-min stabilization period followed by a 1-min recording period.

Although training status was not evaluated in the current study, existing research demonstrates that lnRMSSD data collected daily in the field following only a 1-min stabilization period in elite endurance athletes provided useful information pertaining to training status (Plews *et al.*, 2012, 2013). In addition, a recent case study showed that daily, 55-s lnRMSSD recordings following a brief (i.e. <1 min) stabilization period throughout a collegiate cross-country competitive season related to 8 km race performances (Flatt & Esco, 2014b). Collectively, shortened HRV recording methodology for lnRMSSD assessment in athletes appears promising.

A limitation of the current study is that each participant assumed a supine position for ECG preparation prior to initiating the 5-min stabilization period in line with laboratory ECG procedures. Thus, it is possible that an early lack of stability was undetected due to the 1-min period that it took to prepare each participant for two-lead ECG recording. This has important implications for HRV recording with field tools such as heart rate monitors (Weippert *et al.*, 2010), and smartphone applications (Flatt & Esco, 2013) that require minimal preparation before HRV recording. Therefore, an important next step would be to assess the time course of lnRMSSD stability with HRV field devices that require minimal subject preparation such as those mentioned above. In addition, further research investigating the efficacy of longitudinal lnRMSSD data collected with a 1-min stabilization period for reflecting training status in athletes is needed. Based on the current results demonstrating stabilization of lnRMSSD within the first minute of ECG recording, and in line with existing evidence of its application in the field (Plews *et al.*, 2012, 2013), a 1-min stabilization period prior to an lnRMSSD recording in athletic populations appears sufficient to facilitate a stable condition. It seems that this modified approach to lnRMSSD data collection will yield the same results as the traditional approach of a 5-min recording preceded by a 5-min stabilization period. This may make HRV implementation a more viable option for researchers and sport scientists by reducing the time needed to acquire lnRMSSD data in athletes.

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Conflict of interest

The authors have no conflict of interests

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